

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

COUNTY GOVERNMENT VEHICLE FLEET MANAGEMENT SYSTEM (CVFMS)

Document Version: 1.0

Country: Kenya

Target Users: County Governments

System Type: Web-based Enterprise Fleet Management System

Deployment: Cloud / County Data Centre / Hybrid

Status: Proposed

1. INTRODUCTION

1.1 Purpose

The purpose of the County Government Vehicle Fleet Management System (CVFMS) is to provide an integrated digital platform for planning, acquisition, registration, allocation, dispatch, tracking, maintenance, fuel management, insurance, licensing, inspection, accident management, driver management, utilization monitoring and disposal of county government vehicles and associated fleet assets. The system shall provide county management with accurate, timely and auditable information on fleet operations and expenditure.

1.2 Objectives

- Establish a centralized electronic register of county vehicles.
- Improve vehicle utilization and availability.
- Control vehicle allocation and dispatch.
- Manage official journeys and vehicle requests.
- Manage county drivers and driver compliance.
- Track fuel consumption and expenditure.
- Manage preventive and corrective maintenance.
- Track insurance, inspection, licensing and statutory compliance.
- Record and manage vehicle accidents and incidents.
- Monitor vehicle GPS/location data where tracking devices are installed.
- Provide fleet cost and performance analytics.
- Support vehicle acquisition and disposal processes.
- Prevent unauthorized vehicle use.
- Strengthen accountability and audit trails.
- Integrate fleet information with county finance, HR, procurement, stores and other systems.

2. SYSTEM SCOPE

The CVFMS shall cover the complete vehicle lifecycle: Planning → Acquisition → Registration → Allocation → Dispatch → Utilization → Fuel → Maintenance → Insurance/Compliance → Accident Management → Monitoring → Valuation → Disposal.

Module	Major Functions
Fleet Registry	Vehicle master records and ownership
Vehicle Allocation	Department/station assignment
Vehicle Request	Online vehicle requisitions
Dispatch Management	Trip authorization and dispatch
Journey Management	Trips, destinations and mileage
Driver Management	Driver profiles and compliance
Fuel Management	Fuel issues, consumption and costs
Maintenance	Service schedules and repairs
Workshop Management	Work orders and workshop activities
Insurance	Policies, claims and renewals
Compliance	Inspection, licensing and statutory requirements
Accident Management	Accidents, investigations and claims
GPS/Telematics	Location, mileage and vehicle behaviour
Vehicle Expenses	Operating and lifecycle costs
Procurement	Acquisition and supplier information
Disposal	Valuation, approval and disposal
Stores/Spare Parts	Parts inventory and issuance
Reporting & BI	Dashboards and analytics
Administration	Users, roles, workflows and parameters

Audit	Audit trail and accountability
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3. STAKEHOLDERS

- County Executive: Governor, Deputy Governor, County Executive Committee Members, County Secretary.
- Fleet Administration: Chief Officer/Director responsible for Transport, County Fleet Manager, Fleet Officers, Transport Officers, Vehicle Controllers, Workshop Manager.
- Operational Users: Drivers, departmental vehicle custodians, county officers requesting vehicles, fuel station attendants, mechanics and storekeepers.
- Support Departments: Finance, Supply Chain Management, Human Resource Management, Internal Audit, ICT, Legal, Insurance/Risk Management and County Treasury.
- External Parties: Insurance companies, vehicle inspection providers, garages/workshops, fuel suppliers, GPS/telematics providers, vehicle suppliers, auctioneers and relevant national-government interfaces where legally and technically available.

4. USER ROLES AND ACCESS CONTROL

Role	Key Responsibilities
System Administrator	Configuration and administration
Fleet Manager	Overall fleet management
Transport Officer	Requests, allocation and dispatch
Fleet Officer	Vehicle records and operations
Driver	Assigned vehicle/trip activities
Workshop Manager	Maintenance/workshop
Mechanic	Repair/service activities
Fuel Officer	Fuel transactions
Storekeeper	Spare parts
Finance Officer	Costs and payments
Procurement Officer	Acquisition/contracts
HR Officer	Driver employment information
Auditor	Read-only audit access
Management	Dashboards/reports
Department User	Vehicle requests
Approving Officer	Workflow approvals

Access shall be based on role, department, directorate, vehicle pool, location/station, transaction type and approval authority.

5. FUNCTIONAL REQUIREMENTS

5.1 Fleet Vehicle Registry

The system shall maintain a comprehensive vehicle master register. Capture registration number, chassis/VIN, engine number, make, model, vehicle category/type, year of manufacture, acquisition date/method/cost, current book value, colour, engine capacity, fuel type, transmission, seating/carrying capacity, county asset number, department, directorate, station, custodian, status, GPS device, insurance and inspection/licensing details. Statuses shall include Active, Available, Assigned, On Trip, Under Maintenance, Accident, Grounded, Impounded, Lost/Stolen, Awaiting Disposal, Disposed and Written Off.

5.2 Vehicle Acquisition Management

Record approved fleet requirements, procurement reference, supplier, purchase/contract information, acquisition cost, delivery inspection, registration, asset record, warranty, initial insurance and commissioning date. Acquisition sources may include purchase, lease, hire, donor or transfer.

5.3 Vehicle Allocation Management

Manage allocation to county departments, directorates, sub-counties, stations, projects and authorized officers. Record allocation date, authority, receiving department, custodian, location, expected return date and status.

5.4 Vehicle Request Management

Allow authorized officers to submit electronic vehicle requests containing requester, department, purpose, destination, date/time, expected return, passengers, special requirements, preferred vehicle type, project/activity and supporting documents. Workflow: Request → Supervisor Approval → Transport Review → Vehicle Allocation → Driver Assignment → Dispatch.

5.5 Trip and Journey Management

Maintain trip number, vehicle, driver, passengers, request reference, destination, purpose, departure/return date and time, starting/ending odometer, distance, fuel, authorized route, actual route and status. Calculate distance as ending odometer minus starting odometer.

5.6 Dispatch Management

Before dispatch verify vehicle availability, driver assignment, licence, insurance, inspection, service status, fuel level, odometer, tyres, lights, brakes and safety equipment. Prevent dispatch where mandatory conditions are not met, subject to authorized override.

5.7 Driver Management

Maintain employee number, name, department, designation, phone, licence number/class/expiry, employment status, training, fitness where applicable, assigned vehicle and driving/accident history. Generate compliance alerts.

5.8 Fuel Management

Capture vehicle, driver, date, fuel station, fuel type, quantity, price/litre, total cost, odometer, fuel card, voucher/reference and payment method. Calculate fuel cost and fuel efficiency. Flag abnormal consumption, duplicate transactions, unauthorized locations, quantities beyond configured tank capacity and abnormal efficiency.

5.9 Maintenance Management

Support preventive and corrective maintenance based on mileage, engine hours, time intervals, manufacturer recommendations and county policy. Work orders shall capture vehicle, diagnosis, work, parts, labour, supplier, estimated/actual cost, approvals, completion, odometer and warranty.

5.10 Workshop Management

Manage job cards, vehicle inspection, work allocation, mechanic assignment, spare parts requests/issues, labour, completion, quality inspection and vehicle release. Maintain complete maintenance history.

5.11 Spare Parts and Stores

Manage spare-part catalogue, stock levels, receipts, issues, returns, adjustments, reorder levels, suppliers, compatibility and costs. Link parts to maintenance work orders.

5.12 Tyre Management

Track brand, size, serial number, receipt date, vehicle, position, installation/removal mileage, removal reason, tread and cost. Monitor abnormal tyre consumption.

5.13 Insurance Management

Record insurer, policy number, cover type, dates, premium, vehicle, insured value, claims and renewal status. Generate renewal alerts.

5.14 Licensing and Statutory Compliance

Maintain applicable registration, inspection, licensing, insurance and permit/certificate records. Provide configurable expiry alerts and a compliance dashboard.

5.15 Accident and Incident Management

Record vehicle, driver, date/time, location, description, conditions, other parties, police reference, witnesses, photos, damage, estimated cost, insurance claim, investigation, liability and corrective action.

5.16 GPS and Telematics

Integrate with approved GPS/telematics providers for location, trip history, distance, speed, geofencing, route deviation, idling, excessive speed, unauthorized movement, after-hours movement, engine status and mileage.

5.17 Vehicle Utilization Management

Calculate vehicle utilization, operational days, kilometres, trips, downtime, maintenance downtime, fuel consumption, cost/km, cost/trip, availability and driver utilization.

5.18 Vehicle Cost Management

Track acquisition, insurance, fuel, repairs, maintenance, spare parts, tyres, licensing, inspection, parking, towing, accident and other operating costs. Calculate total operating cost and cost per kilometre.

5.19 Vehicle Disposal Management

Support identification, assessment, recommendation, approval, valuation, disposal method, handover, deregistration and proceeds. Disposal shall follow applicable government and county procedures.

5.20 Financial Integration

Support integration with the County Financial Management System/IFMIS or other authorized financial systems for purchase orders, supplier payments, maintenance/fuel expenditure, asset register, budgets, commitments, payments and disposal proceeds.

5.21 Procurement Integration

Support integration with county procurement systems for procurement plans, requisitions, purchase orders, contracts, suppliers, goods received and invoices.

5.22 HR Integration

Integrate employee master data such as staff number, department, designation, employment status, driver status, transfer and exit.

5.23 Notification and Alert Management

Generate in-system, email, SMS and/or mobile notifications for insurance, inspection, licence, service, maintenance, accident, approval, dispatch, speeding and unauthorized movement events.

5.24 Workflow and Approval Management

Support configurable maker-checker, multi-level approvals, delegation, escalation, rejection, return for correction, approval limits and electronic timestamps.

6. DASHBOARDS AND BUSINESS INTELLIGENCE

Dashboard	Key Indicators
Executive	Total vehicles, active/available, repairs, utilization, fleet value, fuel, maintenance, cost/km, accidents, compliance, disposal
Fleet Manager	Availability, trips, maintenance, fuel, driver performance, GPS alerts, insurance, inspection, licensing, workshop workload
Finance	Fuel, maintenance, cost by vehicle/department, budget vs actual, supplier expenditure

The system shall support configurable analytics, trend analysis, drill-down, filters and export.

7. REPORTING REQUIREMENTS

- Vehicle Register
- Vehicle Allocation Report
- Vehicle Movement Report
- Trip Report
- Driver Report
- Fuel Consumption and Cost Reports
- Maintenance and Workshop Reports
- Spare Parts Report
- Insurance Report
- Compliance Report
- Accident Report
- GPS/Movement Report
- Vehicle Utilization Report
- Cost per Kilometre Report
- Vehicle Lifecycle Cost Report
- Disposal Report
- Fleet Performance Report
- Budget vs Actual Fleet Expenditure
- Audit Trail Report

Reports shall support PDF, Excel and CSV export, subject to role permissions.

8. AUDIT TRAIL

The system shall maintain an immutable audit trail recording user, role, date/time, action, previous value, new value, approval/rejection, comments and permitted technical metadata. Ordinary users shall not be able to edit audit records.

9. SECURITY REQUIREMENTS

- Username/password authentication.
- Single Sign-On where available.
- Multi-factor authentication for privileged/high-risk access.
- Role-Based Access Control.
- Department/location/functional permissions.
- Encryption in transit and at rest.

- Secure password storage and session management.
- Access logging, monitoring, backup and recovery.

10. NON-FUNCTIONAL REQUIREMENTS

Area	Requirement
Performance	Support concurrent county users and large fleet datasets with acceptable transaction response times.
Availability	Target 99.5% availability excluding approved maintenance windows.
Scalability	Support multiple departments, locations, thousands of vehicles and large historical/GPS datasets.
Usability	Responsive web interface, simple navigation, search/filtering and mobile-friendly screens.
Maintainability	Modular architecture, documented APIs, configurable workflows/parameters, logging and version control.
Interoperability	Secure standards-based APIs and integration interfaces.
Accessibility	Follow applicable accessibility and usability standards adopted by the county.

11. MOBILE APPLICATION

- Driver login and assigned vehicle view.
- Vehicle inspection checklist.
- Accept/start/end trip.
- Capture odometer and fuel information.
- Report breakdowns and accidents.
- Upload photos/documents.
- Submit maintenance requests.
- Fleet officer approvals, vehicle search, dispatch, GPS and inspection functions.

12. INTEGRATION ARCHITECTURE

The system shall expose secure APIs and integrate, where approved and technically available, with County SSO, HR, Finance/ERP, SCM, Stores, GPS/telematics, SMS/email and other enterprise platforms. APIs should support REST/JSON, secure authentication, OAuth2/OpenID Connect where applicable, rate limiting, logging, error handling and versioning.

High-level architecture: Presentation Layer (Web Portal, Mobile App, Dashboards) → API/Application Services → Data Services (Transaction DB, Data Warehouse, Document/Audit Store) → Integration Layer/API Gateway → HR, Finance, SCM/Stores, GPS/Telematics and Notification services.

13. MASTER DATA MANAGEMENT

- Vehicle types and makes/models
- Fuel types
- County organizational structure
- Departments/directorates/stations/sub-counties
- Drivers
- Suppliers
- Workshops

- Fuel stations
- Maintenance categories
- Expense categories
- Disposal categories
- Approval levels
- Alert parameters

14. CONFIGURATION MANAGEMENT

Authorized administrators shall configure organizational structures, roles, approval workflows, maintenance intervals, fuel thresholds, speed thresholds, alert periods, vehicle categories, cost centres, locations and business rules. Configuration changes shall be audited.

15. DATA MIGRATION

The implementation shall support migration from Excel/CSV, legacy fleet systems, asset registers and manual registers. Activities shall include profiling, cleansing, mapping, validation, migration and reconciliation, followed by user acceptance.

16. BACKUP AND DISASTER RECOVERY

- Automated database/application/configuration backups.
- Off-site backup for critical data.
- Documented recovery procedures.
- Disaster recovery environment for critical services.
- Recommended RPO: ≤24 hours, preferably ≤1 hour for critical transactional data.
- Recommended RTO: ≤4 hours.

17. DATA RETENTION

The system shall retain historical records in accordance with applicable Kenyan laws, county records-management requirements, financial/audit requirements and approved county policies. Inactive records should be archived rather than physically deleted unless authorized.

18. BUSINESS RULES

Rule	Requirement
BR-001	A vehicle shall not be dispatched if mandatory insurance is expired, unless an authorized policy-based override exists.
BR-002	A vehicle shall not be assigned to an inactive driver.
BR-003	A driver shall not be assigned where the applicable driving licence is invalid or expired.
BR-004	A grounded vehicle shall not be dispatched without authorized override.
BR-005	Every official journey shall have an approved request or authorized operational reference.
BR-006	Every fuel transaction shall identify a vehicle.
BR-007	Odometer readings shall not normally decrease.
BR-008	Maintenance work shall be linked to a vehicle and work order.

BR-009	All approvals shall be recorded in the audit trail.
BR-010	A disposed vehicle shall no longer be available for operational allocation.

19. KEY PERFORMANCE INDICATORS

- Vehicle availability %
- Vehicle utilization %
- Maintenance downtime
- Average kilometres per vehicle
- Cost/km
- Fuel efficiency
- Fuel cost/km
- Maintenance cost/vehicle
- Accident frequency and cost
- Vehicle lifecycle cost
- Fleet age
- Compliance %
- Driver trips/distance/fuel efficiency/speed violations
- Workshop jobs, repair time, downtime, parts consumption and maintenance cost

20. CORE DATA ENTITIES

Vehicle; VehicleType; VehicleAllocation; VehicleAssignment; VehicleRequest; Trip; Journey; Driver; DriverLicence; VehicleInspection; FuelTransaction; FuelCard; MaintenanceSchedule; MaintenanceWorkOrder; Workshop; Mechanic; SparePart; SparePartIssue; Tyre; InsurancePolicy; InsuranceClaim; ComplianceRecord; Accident; GPSDevice; GPSPosition; Geofence; VehicleExpense; Supplier; PurchaseOrder; VehicleAcquisition; VehicleDisposal; VehicleValuation; Department; Directorate; Station; User; Role; Permission; Workflow; Approval; Notification; AuditLog.

21. IMPLEMENTATION PHASES

Phase	Scope
Phase 1 – Foundation	Administration, fleet registry, RBAC, organization, allocation
Phase 2 – Operations	Vehicle requests, dispatch, trips, drivers, mobile app
Phase 3 – Fleet Cost Management	Fuel, maintenance, workshop, spare parts, tyres
Phase 4 – Compliance & Risk	Insurance, inspection, licensing, accidents, compliance
Phase 5 – Digital Fleet Intelligence	GPS, BI, predictive maintenance, optimization and AI analytics
Phase 6 – Lifecycle Management	Acquisition, valuation, disposal and Total Cost of Ownership

22. AI AND ADVANCED ANALYTICS

- Predictive maintenance
- Fuel-consumption anomaly detection
- Vehicle replacement recommendations
- Driver risk scoring
- Accident-risk analysis

- Fleet utilization optimization
- Maintenance cost forecasting
- Vehicle lifecycle cost prediction
- Fraud/anomaly detection

Example: the system may identify a vehicle whose fuel consumption is materially above the configured benchmark for its vehicle class and route profile and generate an investigation alert.

23. FRAUD AND CONTROL MANAGEMENT

- Duplicate fuel transactions
- Fueling outside approved stations
- Fueling without a valid trip/reference
- Excessive fuel quantities
- Odometer anomalies
- Unauthorized vehicle movement
- Ghost trips
- Duplicate maintenance transactions
- Excessive repair costs
- Repeated repairs
- Unauthorized allocation
- Fuel consumption anomalies

24. ACCEPTANCE CRITERIA

- All approved core modules are operational.
- Vehicle records can be created and maintained.
- Vehicles can be allocated and dispatched.
- Requests follow configured approval workflows.
- Drivers can be assigned and validated.
- Trips can be recorded and closed.
- Fuel transactions can be recorded and reconciled.
- Maintenance can be scheduled and tracked.
- Insurance and compliance alerts operate correctly.
- Accidents can be recorded and reported.
- GPS integration operates where applicable.
- Reports reconcile with source transactions.
- RBAC prevents unauthorized access.
- Audit trails are generated.
- Data migration is validated.
- UAT scenarios pass.
- Security testing is completed.
- Performance requirements are demonstrated.
- Backup and recovery procedures are tested.
- County business owners formally approve the system.

25. UAT TEST CATEGORIES

Category	Example
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Fleet Registry	Create/update vehicle
Allocation	Assign vehicle
Request	Submit vehicle request
Approval	Approve/reject request
Dispatch	Dispatch vehicle
Trip	Start/end journey
Driver	Validate licence
Fuel	Record fuel
Maintenance	Create work order
Workshop	Complete repair
Insurance	Renewal alert
Compliance	Expired document
Accident	Record accident
GPS	Track vehicle
Reporting	Generate fleet report
Security	Role restriction
Audit	Verify audit trail
Integration	Finance/HR/GPS
Disposal	Process disposal

26. RECOMMENDED COUNTY FLEET OPERATING WORKFLOW

Vehicle Registry → Allocation → Vehicle Request → Approval → Driver Assignment → Dispatch → Trip → Fuel/GPS/Odometer Capture → Fleet Analytics → Maintenance → Lifecycle Cost Management → Disposal.

The solution should be architected so the Vehicle Fleet module can expand into a broader County Plant & Equipment Management System covering tractors, graders, excavators, rollers, water bowsers, fire engines, ambulances, generators, agricultural machinery and construction equipment.

27. GOVERNANCE AND COMPLIANCE

The final implementation shall be configured to comply with applicable Kenyan public-sector requirements, county policies, approved financial/procurement controls, records-management requirements, data-protection and cybersecurity requirements, and applicable transport/vehicle regulatory requirements. Specific statutory workflows and approval authorities shall be validated with the implementing County Government before configuration.

28. DOCUMENT CONTROL

Field	Value
Document	Software Requirements Specification – County Vehicle Fleet Management System
Version	1.0
Status	Proposed
Owner	County Government / Authorized Implementing Agency
Review Cycle	At major requirements, policy or system changes
Approval	To be completed by authorized County stakeholders